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--Q1 . The finance team wants to identify high-earners who are also working on projects that do not belong to thier own dependants (cross-department projects assignment)  
--unsignedsing correlated sub-queries,list the full name , salary,department number ,and project name of every employee who (1) earns strictly more than average salary of all employees in their own department and (2) is assigned to at least one project whose controlling department differs from the employees own department number

```
SELECT
    E.Fname,
    E.Minit,
    E.Lname,
    E.Salary,
    E.Dno,
    P.Pname
FROM EMPLOYEE E
JOIN WORKS_ON W
    ON E.Ssn = W.Essn
JOIN PROJECT P
    ON W.Pno = P.Pnumber
WHERE
    E.Salary >
    (
        SELECT AVG(E2.Salary)
        FROM EMPLOYEE E2
        WHERE E2.Dno = E.Dno
    )
AND EXISTS
    (
        SELECT *
        FROM PROJECT P2
        WHERE P2.Pnumber = P.Pnumber
        AND P2.Dnum <> E.Dno
    );
```

--Q2.create a view named Young\_Employee that contains all employees who work on more than one project and whose ages are less than the average age of all employees whose first names begin with the letter "J"

```
CREATE VIEW Young_Employe AS
SELECT
    E.Fname,
```

```

E.Minit,
E.Lname,
E.Ssn,
E.Bdate,
CAST((julianday('now') - julianday(Bdate))/365.25 AS INTEGER) AS Age
FROM EMPLOYEE E
JOIN WORKS_ON W
ON E.Ssn = W.Essn
GROUP BY
E.Ssn,
E.Fname,
E.Minit,
E.Lname,
E.Bdate
HAVING
COUNT(DISTINCT W.Pno) > 1
AND
CAST((julianday('now') - julianday(Bdate))/365.25 AS INTEGER)
<
(
SELECT AVG(
CAST((julianday('now') - julianday(Bdate))/365.25 AS INTEGER)
)
FROM EMPLOYEE
WHERE Fname LIKE 'J%'
);

```

```

SELECT * FROM Young_Employee;

```

--Q3.Executive leadership requires a single consolidated analytics report.For each department,compute the following :total headcount , average employees salary rounded to 2 decimal places, highest individual salary,lowest individual salary,salary range(highest minus lowest),total project hours

--contributed by all department employees, the number of distinct projects the department's employees are assigned to and the average hours per project assignment. Include only departments that have at least 2 employees and whose average is above 30,000.Order by average salary descending

```

SELECT
D.Dname,
COUNT(DISTINCT E.Ssn) AS Total_Employees,
ROUND(AVG(E.Salary),2) AS Average_Salary,
MAX(E.Salary) AS Highest_Salary,
MIN(E.Salary) AS Lowest_Salary,
MAX(E.Salary)-MIN(E.Salary) AS Salary_Range,
SUM(W.Hours) AS Total_Project_Hours,
COUNT(DISTINCT W.Pno) AS Number_of_Projects,
ROUND(AVG(W.Hours),2) AS Average_Hours_Per_Project

```

```

FROM DEPARTMENT D
JOIN EMPLOYEE E
ON D.Dnumber = E.Dno
JOIN WORKS_ON W
ON E.Ssn = W.Essn
GROUP BY
    D.Dnumber,
    D.Dname
HAVING
    COUNT(DISTINCT E.Ssn) >= 2
    AND AVG(E.Salary) > 30000
ORDER BY Average_Salary DESC;

```

--Q4.

```

SELECT
    Fname,
    Minit,
    Lname,
    Address
FROM EMPLOYEE
WHERE
    Lname LIKE '____n'
    AND Address LIKE '%Houston%';

```

--Q5.

```

CREATE VIEW dept_manager_repor AS
SELECT
    D.Dname,
    CONCAT(M.Fname, ' ', M.Minit, ' ', M.Lname) AS Manager_Name,
    M.Salary AS Manager_Salary,
    COUNT(E.Ssn) AS Total_Employees,
    ROUND(
        AVG(
            CAST((julianday('now') - julianday(E.Bdate))/365.25 AS INTEGER)
        ),
        2) AS Average_Age,
    SUM(E.Salary) AS Total_Salary_Bill
FROM DEPARTMENT D
JOIN EMPLOYEE M
ON D.Mgr_ssn = M.Ssn
JOIN EMPLOYEE E
ON D.Dnumber = E.Dno
GROUP BY
    D.Dnumber,
    D.Dname,
    Manager_Name,

```

M.Salary;

```
SELECT *  
FROM dept_manager_repor  
WHERE Total_Salary_Bill > 100000  
ORDER BY Total_Salary_Bill DESC;
```